

2024

COMPUTER SCIENCE — HONOURS — PRACTICAL

Paper : DSCC-3P

(Data Structure)

Full Marks : 25

SET - IV

Marks Distribution :

Experiment : 15

Source Code : 08

Algorithm : 04

Output : 03

Viva voce : 05

Laboratory Notebook : 05

Answer *any one* question.

1. Write a program to search for a given target element using Binary Search from an array. If the element is found, return its index; otherwise, return -1.
2. Implement a stack using a linked list with the following operations Push, Pop, Peek, isEmpty and Display.
3. Implement a Double-ended Queue (Deque) using a linked list with functions to insert and delete elements from both the front and rear.

(2)

B(3rd Sm.)-Computer Sc.-H/Pr./DSCC-3P/Set-I-V/CCF

4. Write a program to sort an array using Heap Sort and display the sorted array.
 5. Write a program to find the minimum element in a singly linked list. Implement a function that traverses the list and returns the smallest value present.
 6. Write a program to implement a Doubly Linked List with a search function to find a number in the list and return its position. Implement a function to reverse the doubly linked list.
 7. Write a program to implement a Binary Search Tree (BST) with functions to search for a given element and determine its depth in the tree.
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